



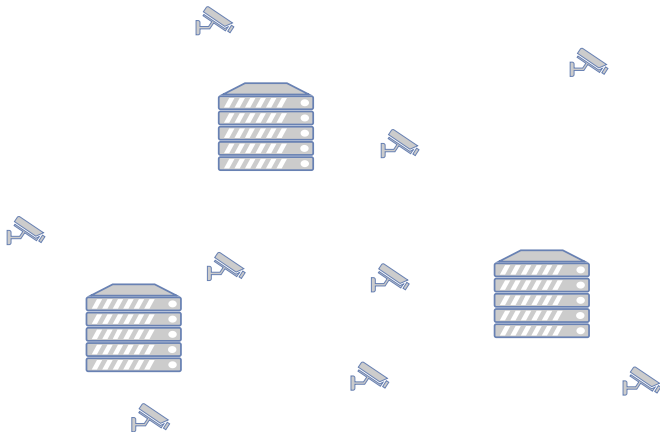
Technische
Universität
Braunschweig



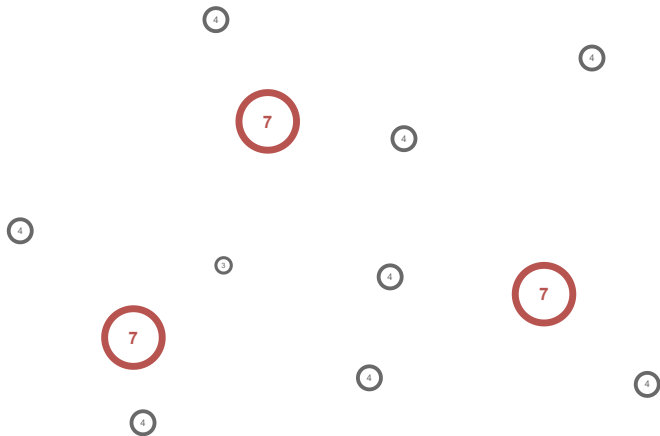
Algorithmische Methoden zu gradbeschränkten Triangulierungen

Fabian Alich, 8. September 2025

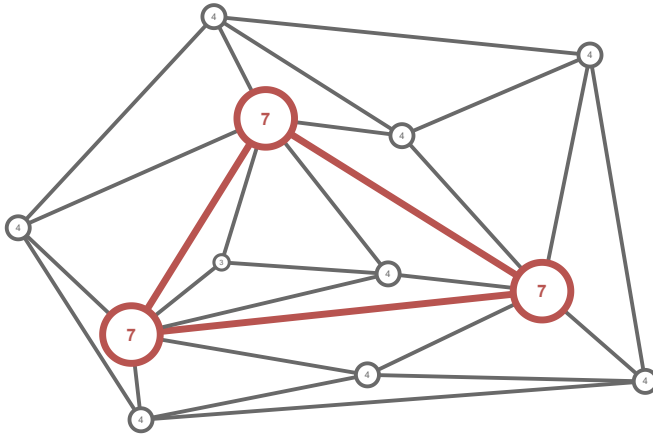
Praktische Relevanz



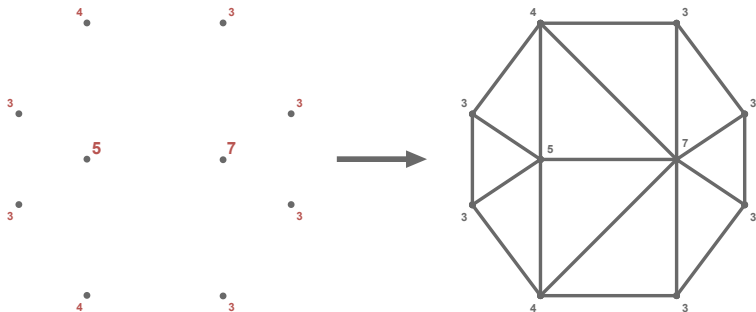
Praktische Relevanz



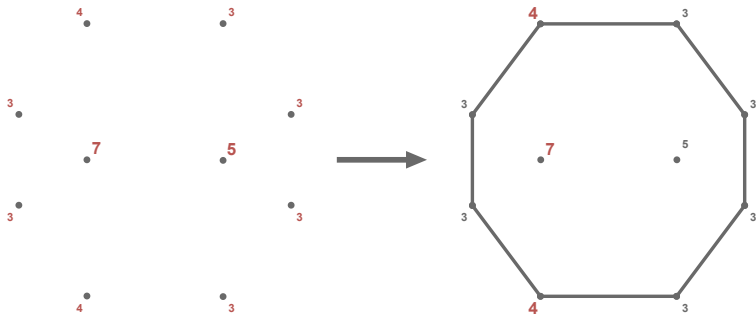
Praktische Relevanz



Ja-Instanz



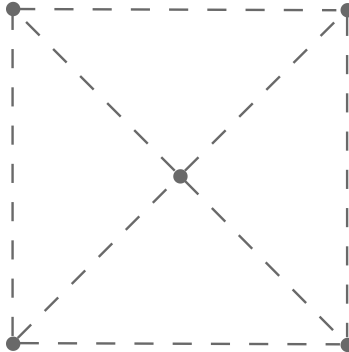
Nein-Instanz



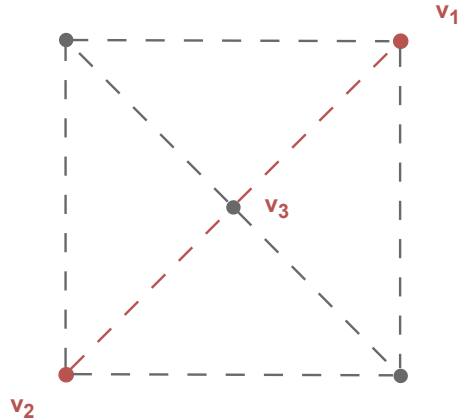
Teil I

Lösungsansätze

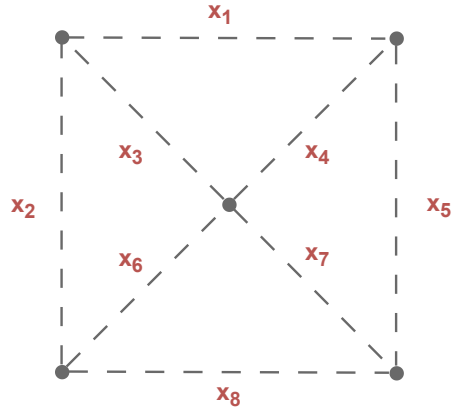
Kantenansatz



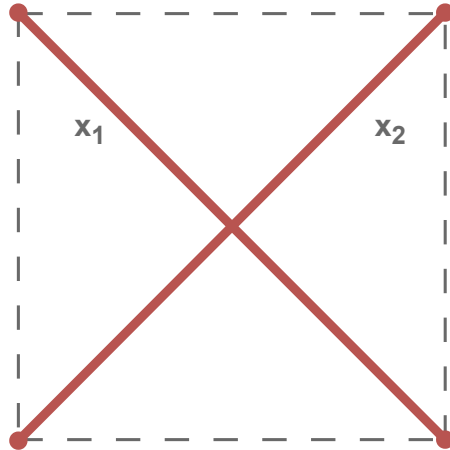
Kantenansatz



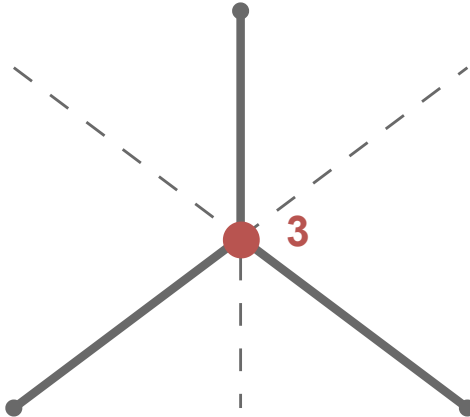
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung (Entscheidungsproblem)

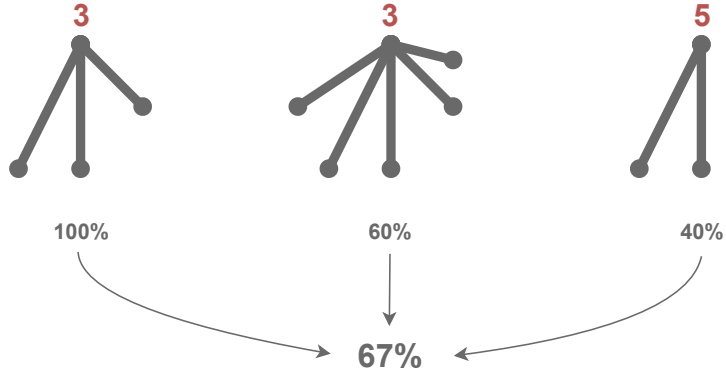


Maximale Kanten (Entscheidungsproblem)

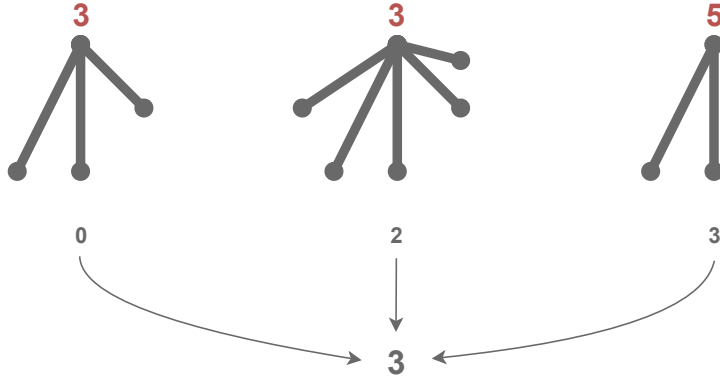
FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

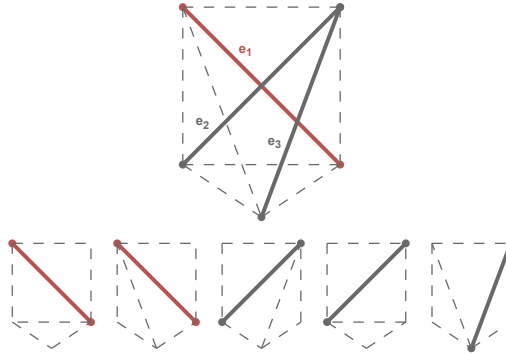
Gradbeschränkung (Optimierungsproblem)



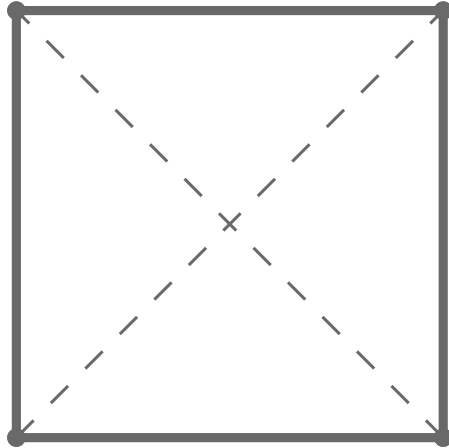
Gradbeschränkung (Optimierungsproblem)



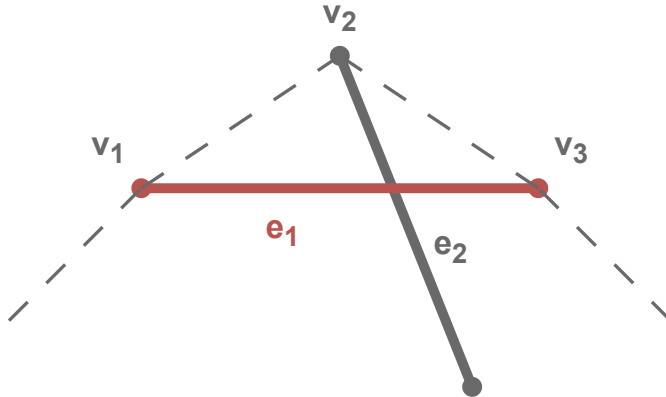
Alternative Kantenbedingung



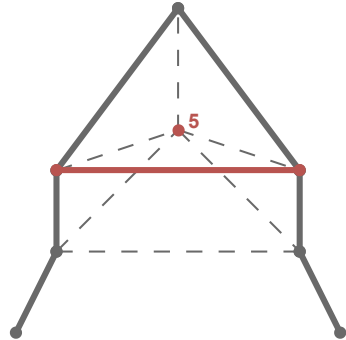
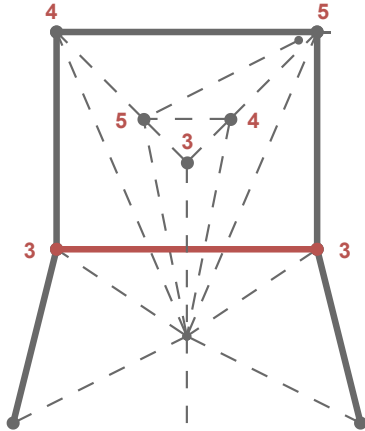
Hülle Fixieren



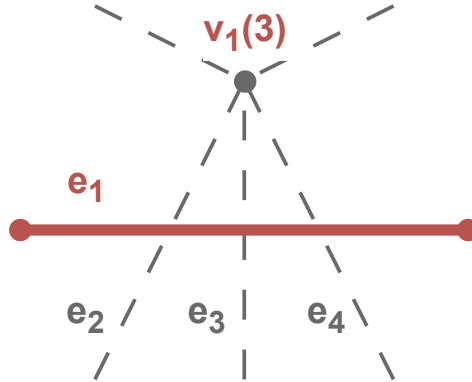
Kanten Fixieren / Ausschließen



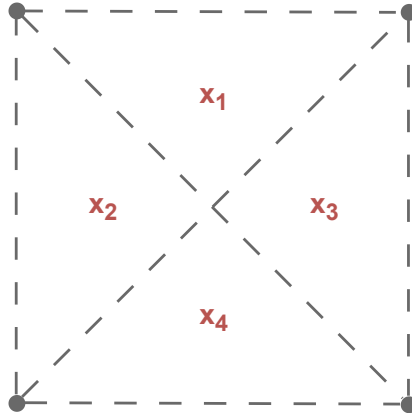
Kanten Ausschließen



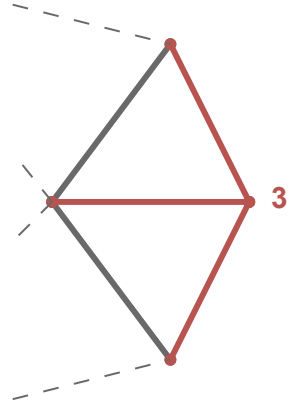
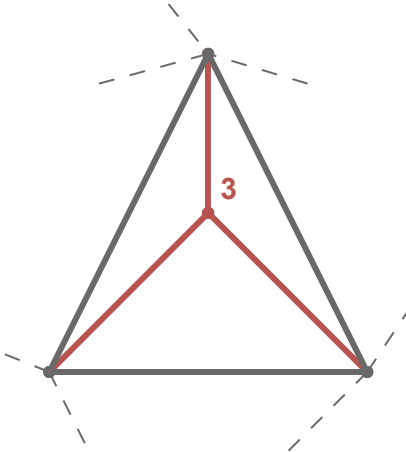
Kanten Ausschließen



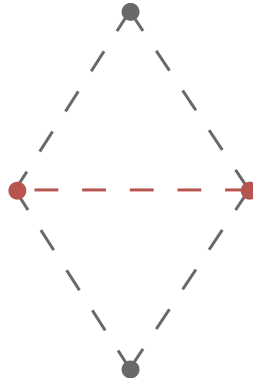
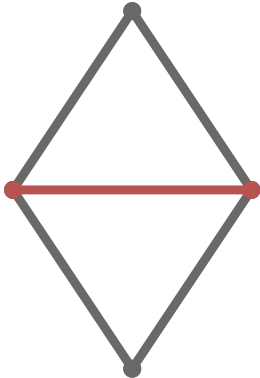
Dreiecksansatz



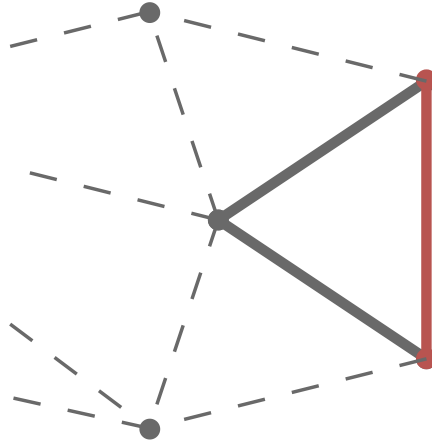
Gradbeschränkung



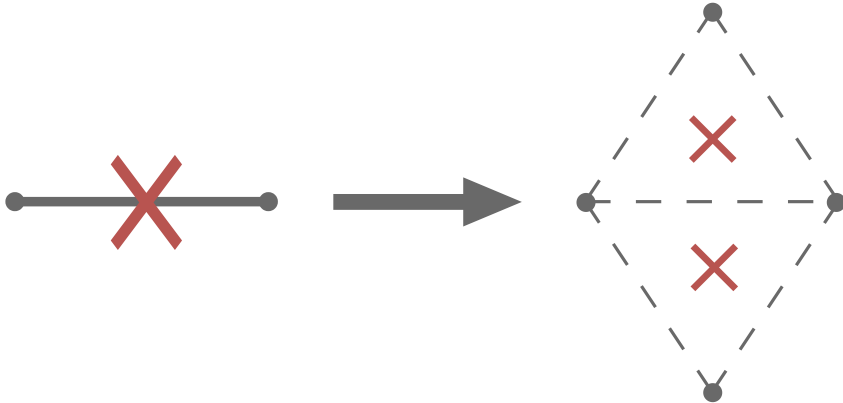
Überschneidungsbedingung



Überschneidungsbedingung



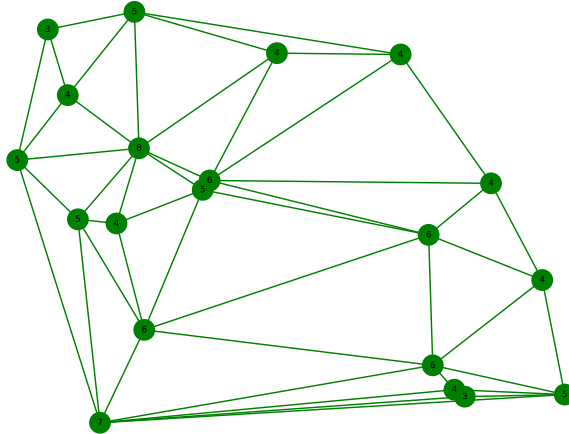
Dreiecke ausschließen



Teil II

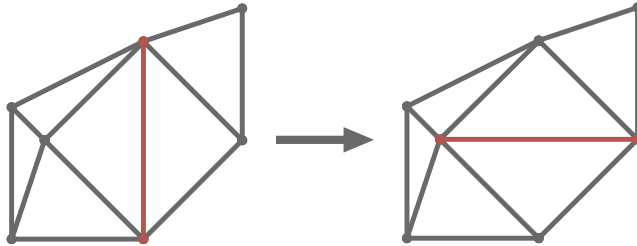
Instanzgenerierung

Delaunay

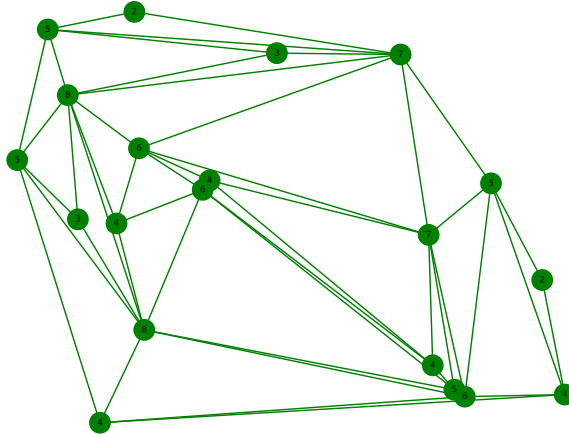


Flip

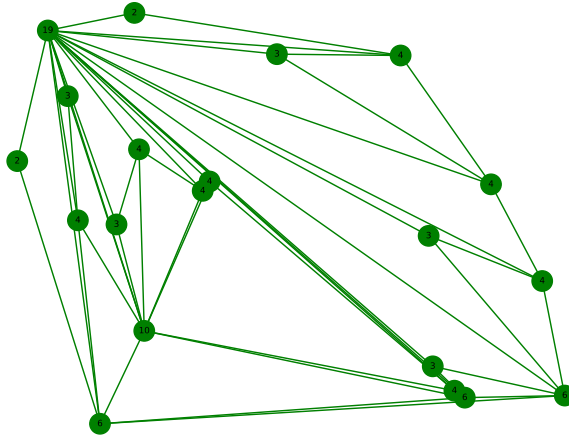
FA: Soll ich das als Animation



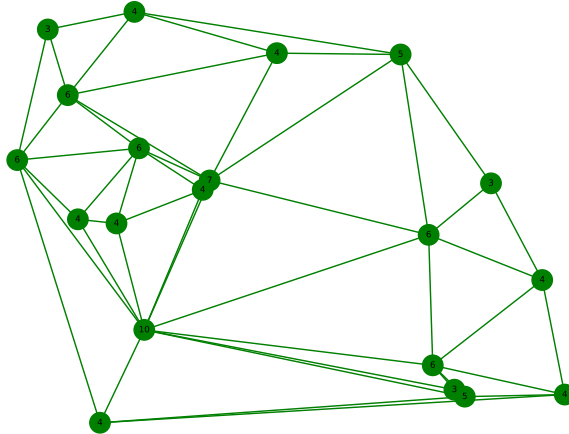
Delaunay-Flips



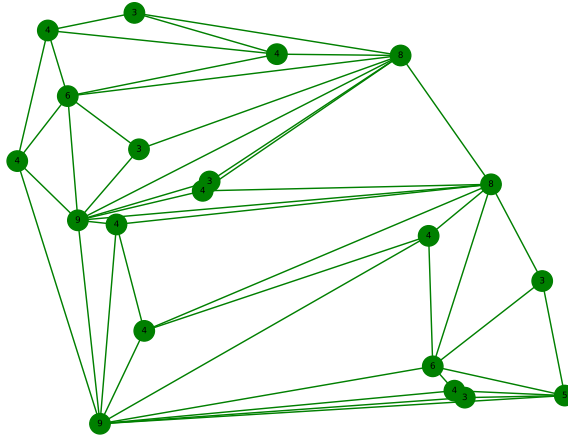
Iterativ



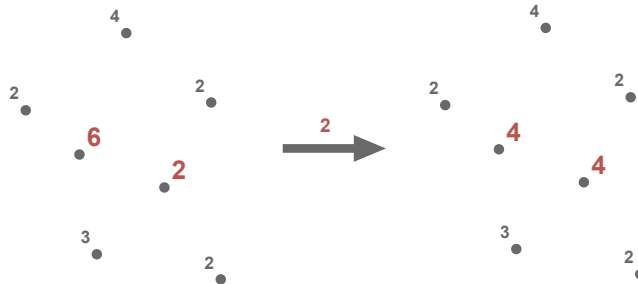
Greedy



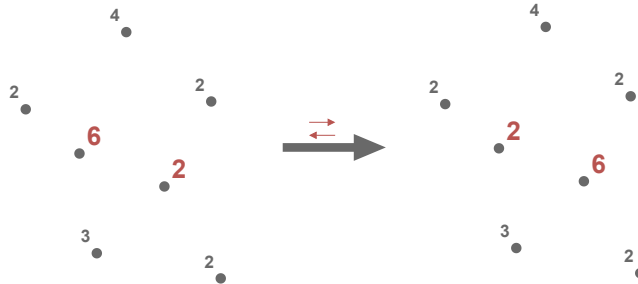
Random



Move



Change



Teil III

Auswertung

Solver



Gurobi

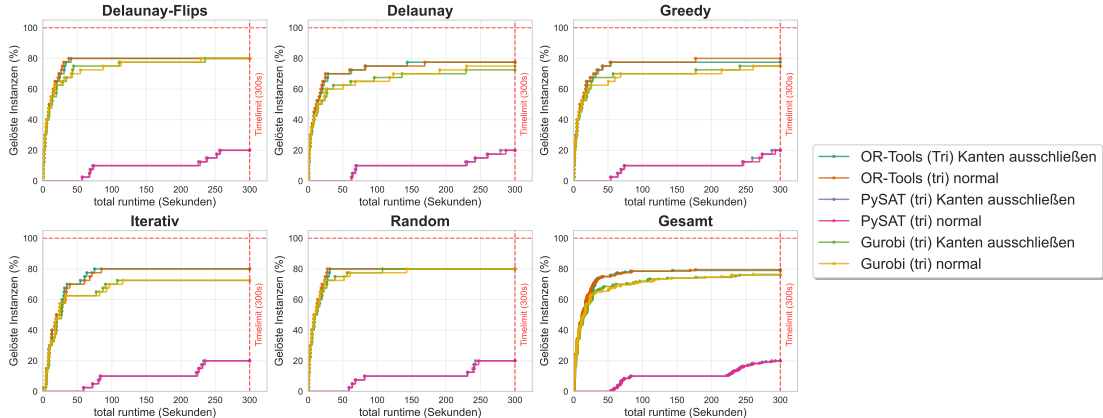


OR-Tools

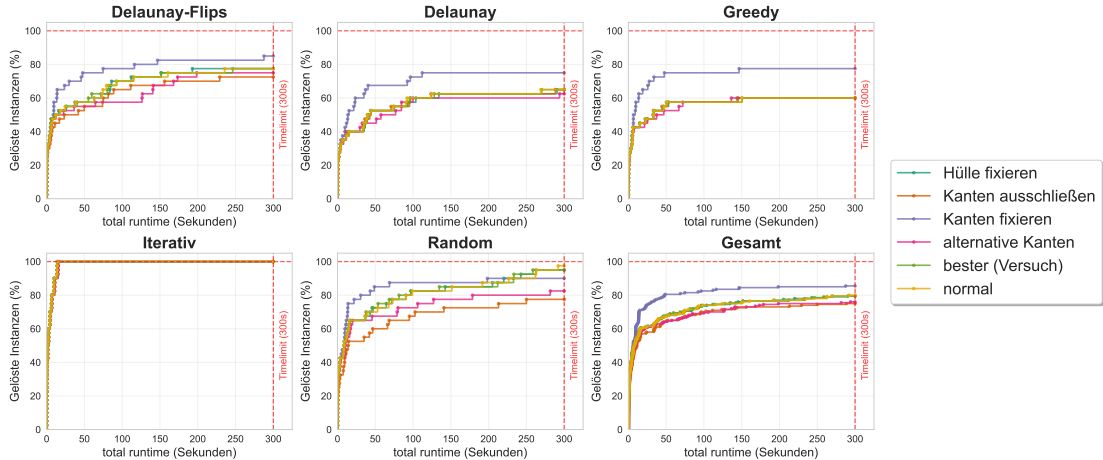


PySAT

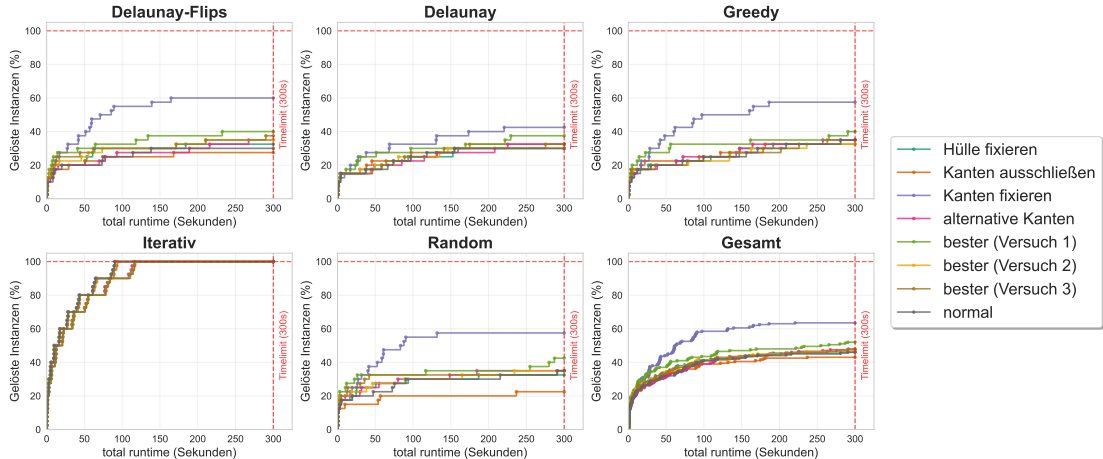
Dreiecke Bedingungen



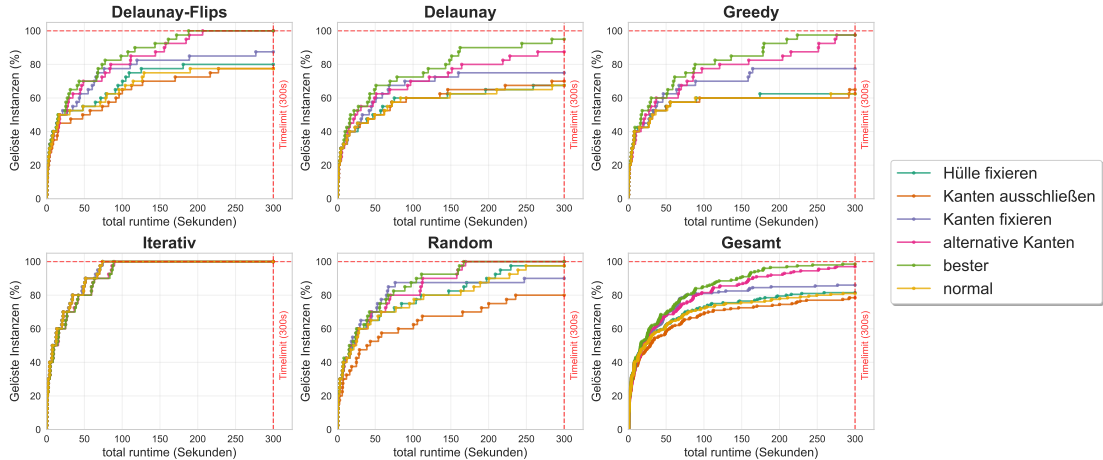
PySAT Bedingungen



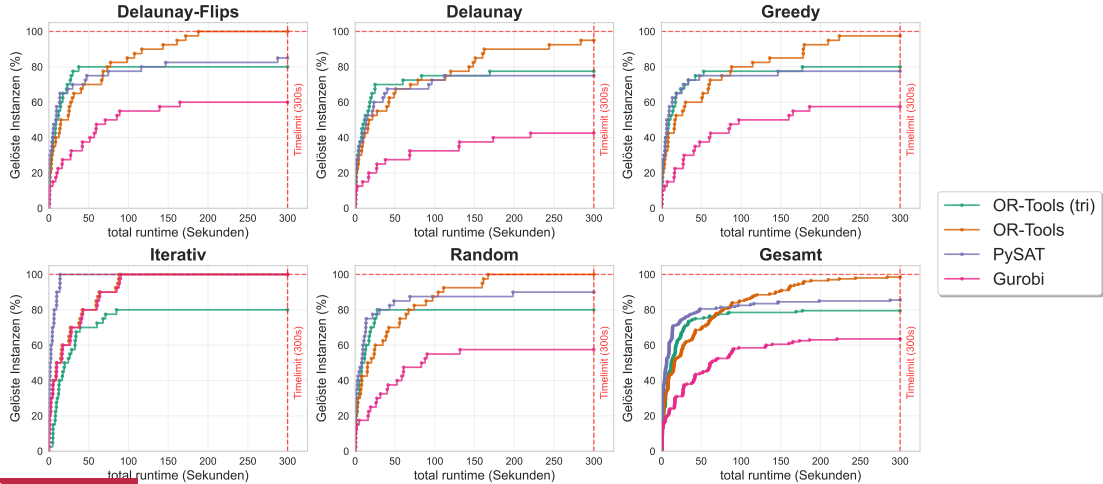
Gurobi Bedingungen



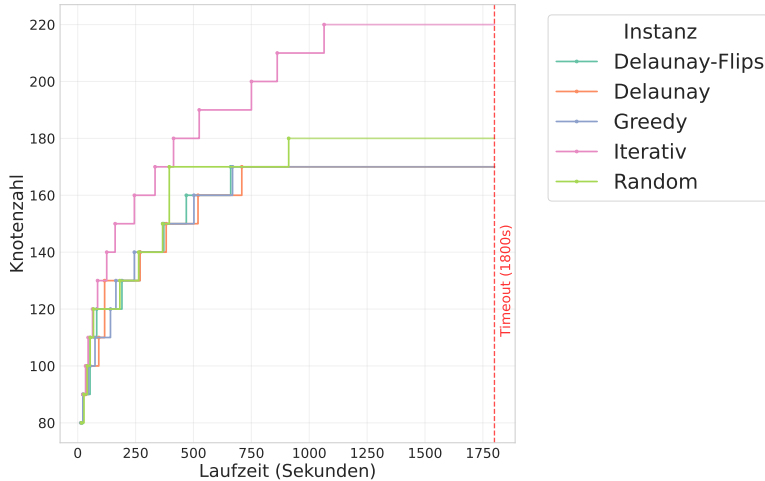
OR-Tools Bedingungen



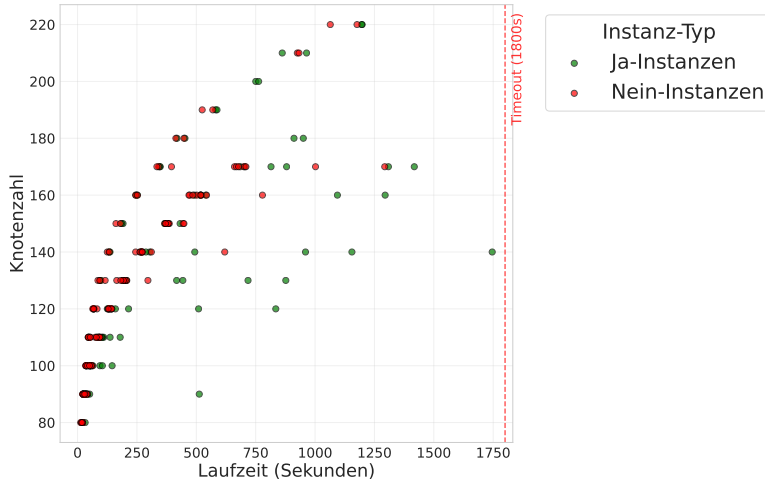
Solver Vergleich



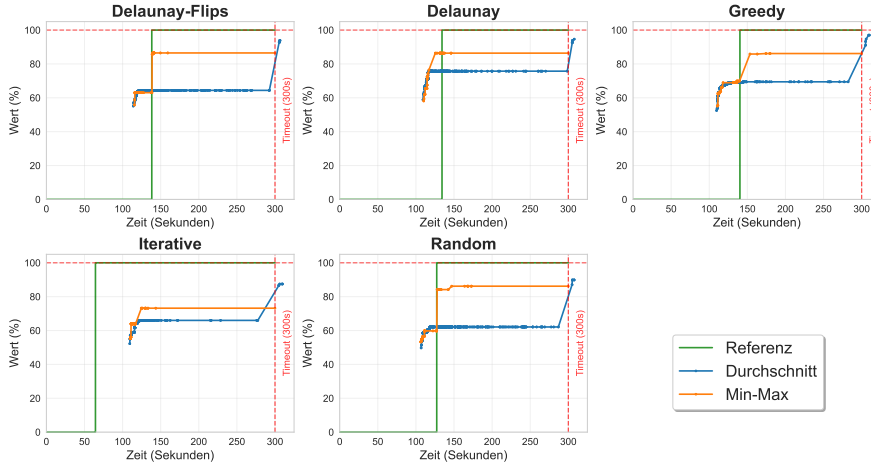
Größerer Timeout



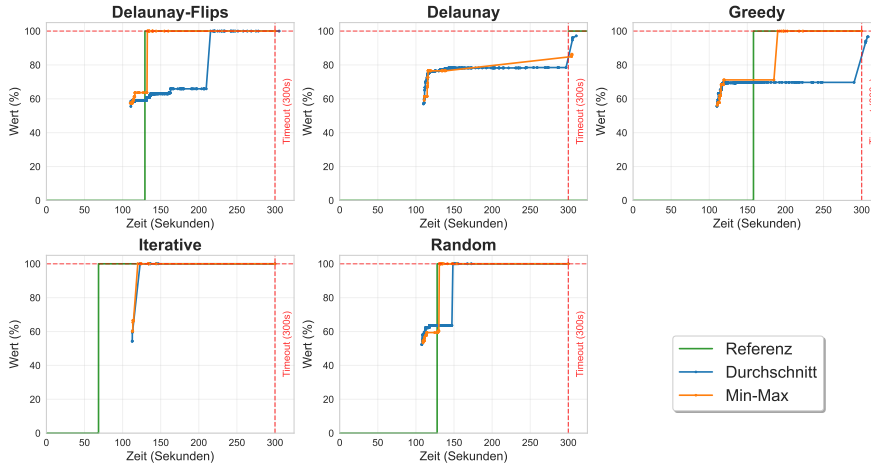
Ja Nein getrennt



Nein Instanz



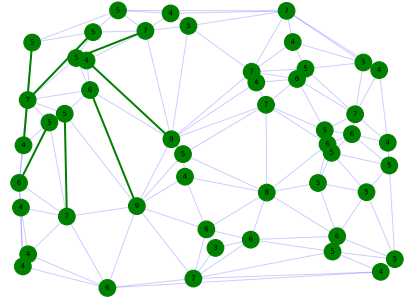
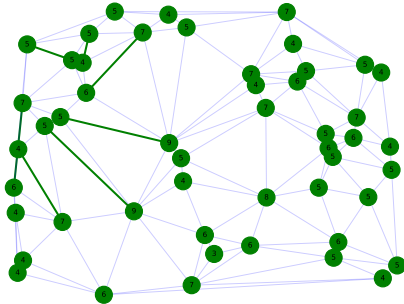
Ja Instanz



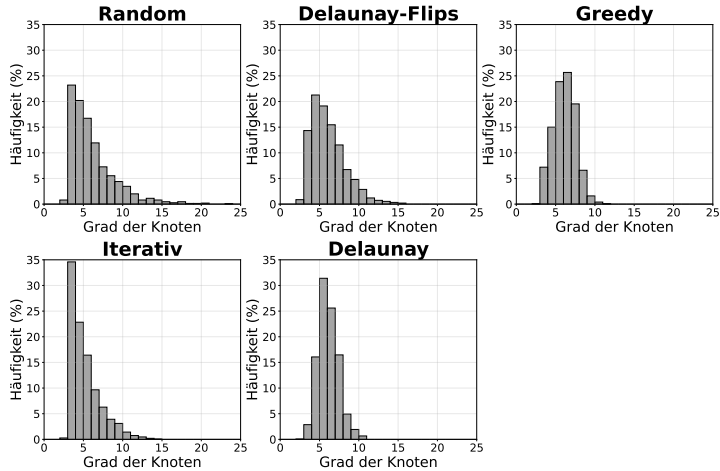
Einordnung

1. Iterativ
2. Random
3. Delaunay-Flips
4. Greedy
5. Delaunay

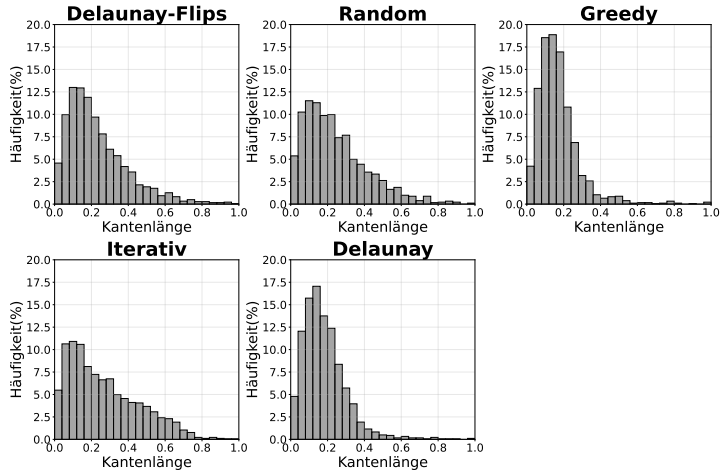
Anzahl Lösungen



Gradverteilung



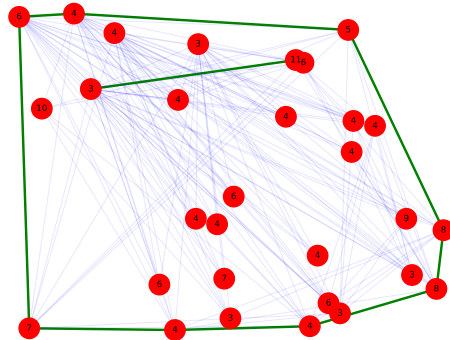
Kantenlängen



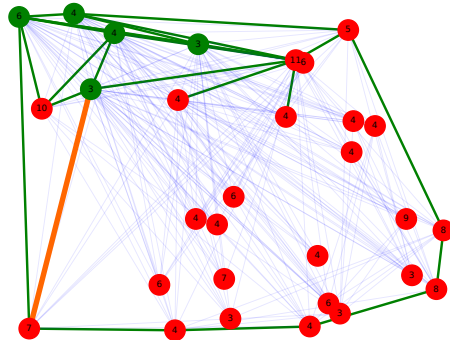
Conflict-driven clause learning

CaDiCaL Simplified Satisfiability Solver

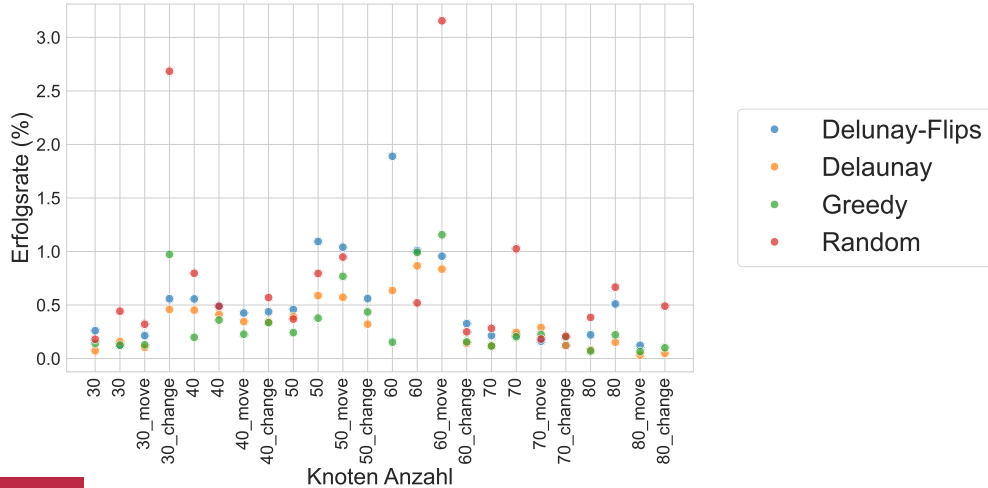
Teilinstanz



Teilinstanz



Propagation



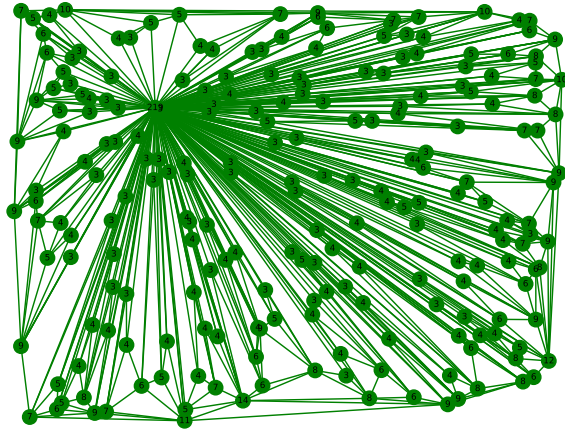
Teil IV

Fazit und Ausblick

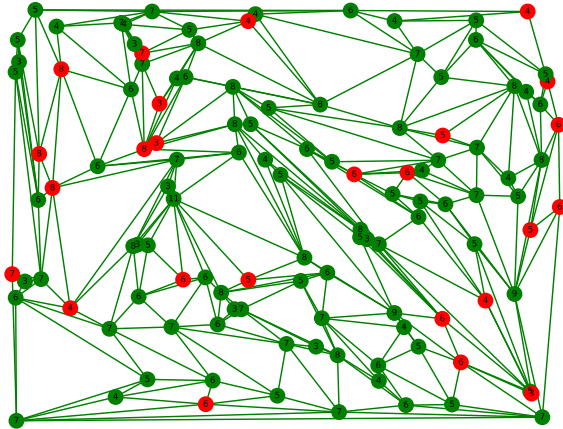
Fazit

OR-Tools Hülle Fixieren Kanten Fixieren alternative Kantenbedingung

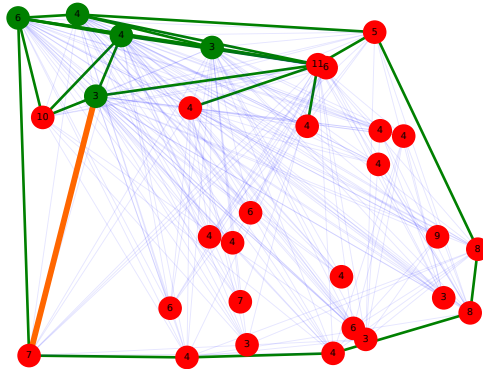
Fazit



Fazit



Fazit



Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierigkeit Instanzen
- Optimierungen Kombinationen
- Propagation erweitern
- normalverteilte Triangulierung
- Problem abwandeln
 - Kanten vorgeben
 - Polygone